

Uncertainty quantification in data management: error assessment for ignition delay times measured in shock tubes

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Abstract

Reliable kinetic modeling and data-driven optimization require experimental databases with quantified uncertainties. Yet >90% of published ignition-delay time (IDT) datasets report none, limiting machine learning and statistical calibration. We present an automated algorithm to reconstruct uncertainty for shock-tube (ST) IDTs when not reported. The method follows GUM/NIST, separating systematic (Type-B) and random components. Systematics are estimated by scoring deviations from an ideal facility/operation using a modified Harrington desirability function applied to key metadata (hardware, conditions, diagnostics, τ -definition). Random uncertainty is obtained by propagating input-state uncertainties through the IDT measurement model. We output combined and expanded uncertainties with standardized, traceable bounds suitable for database integration. The procedure is demonstrated on published datasets and is integrated into the developing kinetics cyber-infrastructure TUMKIN, enabling uncertainty-evaluated IDT corpora for mechanism validation, Bayesian parameter estimation, and data-driven combustion

Keywords: Uncertainty quantification; Data-driven modeling; Ignition delay time; Shock tube; Chemical kinetics database; Systematic uncertainty

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1. Introduction

Process models for reactive systems must rest on detailed mechanisms and experimental targets with quantified uncertainty to remain transferable. In practice, heterogeneous datasets (reactors, boundary conditions) must be assimilated with UQ spanning measurement, facility/systematics, model form, thermochemistry, transport, and numerics. Despite progress in repositories and optimization toolchains uncertainty reporting for ignition-delay times (IDTs) remains sparse and inconsistent, limiting ML/statistical calibration and blurring data-model attribution [1, 2, 3, 4, 5, 6, 7]. Although reporting standards exist [8, 9], adoption is uneven due to the effort required and reproducibility concerns [6].

We address this gap with an automated procedure to assign uncertainty to shock-tube (ST) IDTs when publications omit it. STs enable near-ideal 0-D modeling, yet facility-specific nonidealities introduce systematic biases under nominally identical conditions. Our method, aligned with GUM/NIST, (i) defines a taxonomy and quantitative metrics for facility/condition-dependent systematics; (ii)

estimates the systematic (Type-B) component via a modified Harrington desirability score relative to an “ideal” setup; (iii) evaluates random uncertainty by propagating input-state errors through the IDT model; and (iv) returns combined and expanded uncertainties suitable for database integration. We (v) demonstrate the workflow on published datasets and (vi) quantify the impact of computed vs assumed uncertainties on feasible parameter sets for mechanism assessment.

A submission protocol and open implementation are prepared for integration into TUMKIN, chemical-kinetics cyberinfrastructure under development in the Technical University of Munich.

2. Methodology

We follow the GUM [8], NIST [9], and related guidance [10, 11]. A measurement is complete only when accompanied by a quantitative statement of its uncertainty. Uncertainty reporting includes the expanded uncertainty U together with the coverage factor k used to obtain it, or the combined uncertainty u_c , representing the estimated standard deviation of the result. Uncertainty components are grouped as

Type A (statistical) and Type B (non-statistical). Here, Type A corresponds to the random part $\sigma_r(y)$ and Type B to the systematic part $\sigma_\delta(y)$. The combined standard uncertainty is

The combined standard uncertainty is calculated as

$$u_c(y) = \sqrt{\sigma_\delta(y)^2 + \sigma_r(y)^2} \quad (1)$$

$\sigma_r(y)^2$ is estimated by combining the individual standard uncertainties σ_i of the uncertainty components applying the law of uncertainty propagation. The *systematic error* $\sigma_\delta(y)$ captures device- and condition-specific biases in τ_{ign} measurements in shock tubes. We assume an ideal 0-D description (chemistry behind the shock), and quantify departures from this idealization by: (i) identifying dominant facility/operating factors; (ii) constructing factor-wise estimators; (iii) defining a quasi-ideal reference (baseline systematic error 5%); (iv) computing deviation metrics (via a modified Harrington desirability score); and (v) aggregating incremental penalties to obtain $\sigma_\delta(y)$. This treats $\sigma_\delta(y)$ as an offset/zero-setting error. Because rigorous assessments of systematic error in τ_{ign} are scarce, our primary focus is to operationalize this component.

The expanded uncertainty is

$$U = k u_c(y) \quad (2)$$

and we report the coverage interval as $Y = y \pm U$. Consistent with international practice, we use $k = 2$, giving $\approx 95\%$ coverage.

3. The systematic uncertainty at the shock tube measurements of τ_{ign}

A purely statistical/ML treatment of shock-tube systematics is not feasible at present: (i) true multi-lab repeats at identical (T,p, ϕ) are scarce; (ii) sensitivities of individual impacting factors (IFs) are not identifiable from available data; and (iii) no validated stochastic model for systematic error exists at scale. Following GUM/NIST, we therefore treat the systematic component as epistemic and estimate it via Type-B reasoning (expert knowledge, design/operation constraints, literature).

Typical IDT shock tubes probe post-reflected-shock states of ≈ 600 –3000 K and pressures from sub-atm to hundreds of atm (design-dependent). For simulation, kinetic modelers adopt an ideal 0-D reactor behind an instantaneously formed planar reflected shock (uniform, quiescent gas; isochoric or isobaric). Under this idealization, τ_{ign} is chemistry-controlled; departures from ideality map directly to the systematic error $\sigma_\delta(y)$.

3.1 Components of systematic uncertainty: Unavoidable non-idealities

The systematic error is installation-conditioned. We treat it as Type-B (epistemic) and parameterize it by a relative standard deviation δ (percentage points). Figure 1 (network diagram) summarizes facility/operation factors and their links to $\sigma_\delta(y)$. Primary sources of non-ideality [12, 11, 13] [14, 15, 16, 17] [18, 19, 20]: Boundary layer (dominant): side-wall growth $\rightarrow$ non-planar flow, incident-shock attenuation, reflected-shock interaction, enhanced heat transfer; perturbs T_5, P_5 and violates homogeneity; Finite shock formation/diaphragm opening: non-instantaneous planarity; imperfect diaphragms degrade constant-V/p assumptions; Contact surface/reflected shock interaction: mixing/stratification near the end section; Pre-ignition heat release / pressure rise: chemistry alters the intended initial state; Spatially non-uniform, multi-stage ignition: position-dependent τ ; Impurities / radical seeding: adsorbates, outgassing; shortens τ for radical-sensitive fuels (typically smaller than pressure-rise effects).

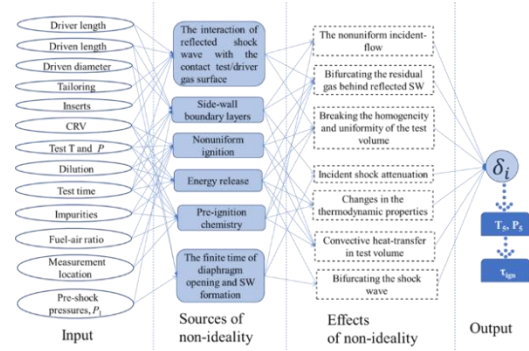


Fig. 1: Network for evaluation of the systematic uncertainty term.

The non-idealities are sensitive to the design, operation conditions and mitigations [12, 11, 21, 22, 23, 24, 13, 25] [14, 26, 17]: Larger diameter/length reduce boundary-layer and contact-surface effects; a longer driver delays the return expansion wave; higher p and dilution temper non-idealities. Reported mitigations include driver inserts (limit attenuation), driver tailoring, diaphragmless operation, constrained-reaction-volume (CRV) strategies, and high-pressure aerosol ST designs.

Disturbances grow with test time; at low T, low p, and lean ϕ , τ can be tens of ms, amplifying boundary-layer/contact-surface impacts. To limit spatial bias, many labs place sensors 1–2 cm from the end wall.

3.2 Target metrics for the “ideal case” and baseline systematic error δ^0

Guided by prior analyses of shock-tube non-idealities, we select 14 input/operating factors (IFs) (Fig. 1, Table 1) that define proximity to the 0-D “ideal case.” A quasi-ideal reference is specified by fixing each factor to its target metric μ_i^0 , Table 1, and assigning a baseline systematic error $\delta^0 = 5\%$ at 95% coverage. This baseline represents the irreducible risk under best-case conditions; deviations from μ_i^0 are treated separately.

Table 1: Impacting factors (IF) and target metrics for the “ideal case” $\delta_0 = 5\%$ (95% confidence).

Category	#	Impacting Factor (IF)	Target metric, μ_i^0
Test equipment	1	Driver length	≥ 3.0 m
	2	Driven length	≥ 8.0 m
	3	Driven inner diameter	≥ 10.0 cm
	4	Tailoring	yes
	5	Inserts	yes
	6	CRV	yes
Operating conditions	7	Reflected-shock Temperature, T_5	1000 - 1600 K
	8	Reflected-shock Pressure, P_5	5 - 20 atm
	9	Valid test time, $\Delta t_{\text{valid}}^*$	50 - 500 μs
	10	Dilution (Ar/He/...)***	yes
	11	Impurities	no
	12	Measurement location	≤ 2 cm (from the end of the tube)
	13	Fuel-oxidizer equivalence ratio	≥ 0.3
	14	Pre-shock pressures, P_1	≥ 10 Torr

* Δt_{valid} time window before contact-surface/expansion-wave arrival or significant boundary-layer growth invalidates 0-D assumptions.
 ** “Yes” indicates purposeful dilution to suppress non-ideal gas dynamics and heat-release effects (typical practice is high inert fraction; specify per system if needed).

3.3 Type-B estimation of systematic uncertainty

We calibrate the systematic error against a quasi-ideal reference through departures of IFs from its reference target. Assuming the independence of all other IFs these departures are mapped to incremental penalties via a smooth “undesirability” transform, and aggregated as

$$\delta = \delta^0 + \sum_{i=1}^N \delta_i = \delta^0 (1 + \sum_{i=1}^N \xi_i), \quad \delta_i = \delta^0 \xi_i, \quad i = 1, \dots, N \quad (3)$$

where N is the number of IFs, and $\xi_i \in [0,1]$ quantifies the penalty induced by the deviation of μ_i from the ideal μ_i^0 .

To define ξ_i , we use a two-sided, smaller-is-better modification of Harrington’s desirability concept [27] [28] retaining low sensitivity near the ideal point and saturation for large deviations.

$$\xi_i = 1 - e^{-(x_i^2)} \quad (4)$$

$$x_i = (\mu_i^0 - \mu_i) / \mu_i^0 \quad (5)$$

For binary IFs (e.g., tailoring, inserts, CRV, dilution, impurities), we map an unfavorable setting to the conservative lower edge of the “satisfactory”

band, i.e. $\xi_i = e^{-1} \cong 0.37$, if “no” and to $\xi_i = 0$ if “yes” for IF_{4,5,6,10,13}. For IF₁₁ it is otherwise.

To avoid identifiability and double counting, we keep IF₉ (test time) in the systematic term and exclude T_5 and P_5 from δ their contributions enter the Type-A propagation. IF₁₄ (P_1) is dropped due to limited quantitative evidence.

3.4 Worked example for δ

We applied the systematic uncertainty calculation for the experimental setup of S. Saxena et al., [29]. Results were obtained for the mixture 3.5% C₂H₄–3.5% O₂–93% Ar, $p_5 = 2.0 \div 2.5$ atm. Systematic expanded relative uncertainty δ is between 9.2–14.2%.

4. Shock tube experiments: random uncertainty

The measured value should be reported along with an estimate of the total combined standard uncertainty, sum of systematic error and random error. In principle, random error is obtained from repeat measurements; when repeats are impractical, we treat the measurand as function of parameters and evaluate the random part by the law of uncertainty propagation. The standard deviation of a measurand is calculated as the positive square root of the combined variance, $u_c^2(y)$. Assuming independence of input parameters [8]:

$$u_c^2(y) = \sum_{i=1}^N \left(\frac{\partial f}{\partial z_i} \right)^2 u^2(z_i) + 2 \sum_{i=1}^{N-1} \sum_{j=i+1}^N \frac{\partial f}{\partial z_i} \frac{\partial f}{\partial z_j} u(z_i) u(z_j) r(z_i, z_j) \quad (6)$$

where z_i are parameters of the function f ; $u^2(z_i)$ is the standard uncertainty of parameters; $r(z_i, z_j)$ is the estimated correlation coefficient.

To describe the set of measured IDT data we used the function proposed by A. Lifshitz [10]:

$$\tau_{ign} = f(z_1, A, B, C, D) = A \exp\left(\frac{B}{z_1}\right) z_2^C z_3^D \quad (7)$$

Here $z_1 = T_5$, $z_2 = P_5$, and $z_3 = \varphi$. A, B, C, D – parameters of approximation. We apply eqs. (4) and (5) under assumptions A, B, C, D, T_5, P_5 , and φ are not correlated parameters and terms with parameter correlations can be neglected.

The standard uncertainty of input parameters $u^2(z_i)$, i.e., manufacture’s specifications, calibration, and other certificates should be reported in a paper. If it is not so it can be evaluated from the available information, analogy, etc. Uncertainties of the approximation parameters $u^2(A)$, $u^2(B)$, $u^2(C)$, $u^2(D)$ are calculated by approximation with standard procedure. Assuming normal distribution of x_i and 95% of confidence interval one can interpret U_i as uncertainty interval

$$U(z_i) = 2\sqrt{u_c^2(z_i)} \quad (8)$$

and the standard uncertainty of parameters can be calculated as

$$u^2(z_i) = \frac{U^2(z_i)}{4} \quad (9)$$

The standard uncertainty of a set of IDT data can be calculated from the sum of systematic and random uncertainties, and eq. (6)

$$u_{ci}(\tau_{ign}) = \sqrt{\sigma_{\delta_i}^2 + \sigma_i^2(\tau_{ign})} \quad (10)$$

$\sigma_{\delta_i}^2$ is the standard systematic deviation. As we interpret δ as normal, 95% expanded, we obtain:

$$\sigma_{\delta_i}^2 = \left[\tau_{ign} * \left(\frac{\delta_{i/2}}{100} \right) \right]^2 \quad (11)$$

The final extended uncertainty of the measured set will be calculated as (in the case of normal distribution of χ_i and 95% of confidence interval):

$$U_i(\tau_{ign}) = 2(u_{ci}(\tau_{ign})) \quad (12)$$

Finally, data accepted for the further analysis and model testing, fitting and optimization are reported as: Final uncertainty interval

$$Y^{exp} = \tau_{ign}^m \pm U(\tau_{ign}) \quad (13)$$

where τ_{ign}^m is measured value of the IDT.

4.1 Worked example for
 $\sigma_i^2(\tau_{ign}), u_{ci}^2, U(\tau_{ign})$

For the same setup [29], we calculate now the random error applying described procedure for reported conditions: $T_5 \pm 1\%$ and $p_5 \pm 1.5\%$; uncertainty in test gas composition was 0.5%. The measured set of data [29] has been approximated with the function:

$$\tau_{ign} = 7.5 \cdot 10^{-3} \exp\left(\frac{16400}{T_5}\right) P_5^{1.91} \varphi^{-3.38}, \mu s \quad (14)$$

The coefficients of eq. (16) were calculated with the uncertainty intervals: $A = 7.5 \cdot 10^{-3} \pm 4.6 \cdot 10^{-5}$
 $B = 1.6 \cdot 10^4 \pm 86.0$, $C = 1.91 \pm 0.07$, $D = -3.38 \pm 0.05$
Figure 2 shows calculated uncertainty intervals. For this dataset, the expanded combined uncertainty is 14.9–25.5% ($k=2$), with the largest relative errors occurring for the long/short IDTs (~ 2100 – $1020 \mu s$). To enable flexible IDT reporting with uncertainties the shock-tube submission should include (machine-

and human-readable): driver length, driven length, driven diameter, test time, tailoring, inserts, CRV, dilution, impurities, and measurement location.

5. Conclusions

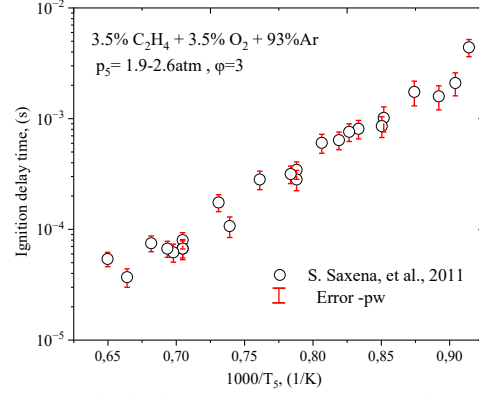


Fig. 2. Calculated uncertainty intervals for τ_{ign} data set from [29]

We present an automated, GUM/NIST-consistent method to assign uncertainties to shock-tube IDTs, including cases where publications reported none.

The algorithm separates uncertainty into Type-B, systematic, and Type-A, random, and returns the combined standard and expanded uncertainty ($\sim 95\%$ coverage).

Systematics are quantified as departures from a quasi-ideal 0-D case: a baseline 5% is assigned under best-case conditions, and deviations of defined impacting factors are mapped via a modified Harrington desirability to penalties that aggregate into δ .

Random uncertainty is obtained by propagating input uncertainties through the IDT measurement model.

The workflow enables standardized reporting, supports database ingestion, data-consistency analysis, and Bayesian/mechanism optimization, offering a reproducible template for experimentalists and data curators.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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